ModelGate

Intelligent AI Routing — Built from Your Contracts

One line changed. Millions of premium calls rerouted.

FastAPI + Python · Next.js + TypeScript · Docker

Team: Agents Assemble

Aaryan Kapoor - Lead Architect and AI Engineer
Pradyumna Kumar - Platform Architect and Frontend
Danny Tran - Design and Presentation Lead

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The Assurant Way

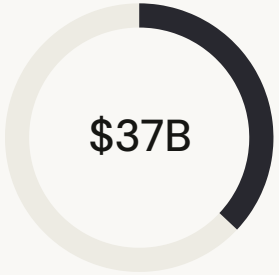
"Helping people thrive in a connected world."

Energy Difference

More energy consumed by premium models vs small models per query

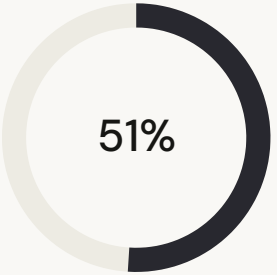
Wasted Spend

Of enterprise AI inference spend wasted on over-provisioned models



Enterprise Spend

Enterprise GenAI spend in 2025 up 3.2× year-over-year



ROI Measurement

Of organizations can actually measure their AI ROI

> Global data center consumption: Premium models: ~39 Wh/query Small models: ~0.22 Wh/query . A 180× energy difference per call.. *IEA Energy Report (2025)*

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The Problem Is Universal

Every layer of the AI stack is affected.

Affected Party	The Problem Today	Consequence
Developers & Product Teams	No time to evaluate 200+ models. Pick one **premium option** and never revisit it.	Overpay by 10–30×
Enterprise Customers	Contract constraints manually translated into config — or **silently ignored**. Compliance risk.	Legal exposure
End Users	"What is your return policy?" routed to a **slow premium model**. Waits 2 seconds for a simple answer.	Poor experience
AI Data Centers	**Unnecessary load** on premium GPU clusters from queries that a small model would answer perfectly.	Wasted capacity

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The Fix

One line of code. We handle everything else.

Before — Every Team Today

```
# Direct provider call
from openai import OpenAI
client = OpenAI(
    api_key="sk-...",
    base_url="https://api.openai.com/v1"
)
# Same model. Every request. Forever.
```

After — With ModelGate

```
# Same code. Different URL.
from openai import OpenAI
client = OpenAI(
    api_key="",
    base_url="https://assurant.com/samsung/v1"
)
# Right model. Every request. Auto.
```

API Compatible

Full OpenAI spec — no SDK changes, no rewrites.

Policy Enforced

Contract constraints become routing rules, automatically.

Always Optimal

Cheapest model that satisfies every constraint, per request.

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Who It Serves

Three Personas. One Solution. Simultaneous benefit.

Developer / Product Team

"I want the best AI model. I don't have the time to evaluate every new model"

Enterprise Compliance

"Our contracts ban certain providers. I'm not sure engineers will remember all the time"

Operations / Platform

"I have no idea what our AI is costing us or which models we're using"

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System Architecture

Two phases. Fully automated. No ongoing effort.

Phase 1: Onboarding

≈ 30 seconds

- Upload contracts, SLAs, etc.
- LLM reads and extracts constraints
- Structured Customer AI Profile generated
- OpenAI-compatible endpoint goes live

Phase 2: Runtime Routing

~50ms overhead

- Prompt received and classified
- Heuristic filters applied
- Models scored and ranked
- Best model selected and response routed

Simple

Ex. Models: GPT 5.4 nano · Gemini 3.1 Flash Lite · Haiku 4.5

Use: FAQ, greetings, status checks

Medium

Ex. Models: GPT 5.4 mini · Sonnet 4.6 · Grok 4.1 Fast

Use: Summarize, explain, troubleshoot

Complex

Ex. Models: GPT 5.4 · Opus 4.6 · Gemini 3.1 Pro · Grok 4.20

Use: Multi-doc reasoning, code gen, legal

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Live Demo 🚀

Five steps. One contract. Immediate routing.

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Evidence

ModelGate Live Results

<30s

Onboarding

Contract → endpoint

~50ms

Classification

ModelGate (Fine-tuned)

85%

MMLU Subset

59%

Savings vs Always-Premium

> "Smart routing delivers 85–98% cost reduction while maintaining quality indistinguishable from always-premium."
>
> Up to 98% cost reduction while matching GPT-4 quality by cascading through cheaper models. *FrugalGPT (Stanford, 2023)*
>
> >85% cost reduction at 95% of GPT-4 performance using a trained routing model. *RouteLLM (LMSYS, ICLR 2025)*
>
> Global data center consumption: 415 TWh in 2024. Projected to exceed 1,000 TWh by 2026. *IEA Energy Report (2025)*

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GPT-5.4 vs. ModelGate Router

Subject	GPT-5.4-High	Router	Delta
Elementary Math	100%	80%	-20%
US History	90%	90%	0
Computer Science	90%	80%	-10%
College Biology	100%	100%	0
Abstract Algebra	60%	70%	+10%
Formal Logic	100%	90%	-10%
Overall	**90%**	**85%**	**-5%**

Cost comparison: $0.0095 (router) vs $0.023 (always GPT-5.4) — projected $1.58/mo vs $3.83/mo at 10k requests

Takeaway: 5pp accuracy trade-off. **59% cheaper**. Hard questions routed correctly.

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Model Fine-tuned

83.3%
Accuracy

50ms
Latency

+71.4%
Medium Fix

3.2x
Faster

Tier	Stock	Fine-Tuned	Delta
Simple	87.9%	81.8%	-6.1%
Medium	**14.3%**	**85.7%**	**+71.4%**
Complex	100%	85.7%	-14.3%
Overall	**70.4%**	**83.3%**	**+13.0%**

- **Method:** GRPO (reinforcement learning) via Unsloth + TRL, LoRA rank 32 (2.3% of params)
- **Base model:** Arch-Router-1.5B (Qwen2.5-1.5B) → GGUF Q8_0 quantized to 1.6 GB
- **Training:** 150 steps, 172 labeled prompts, RTX 3080 Laptop 8GB, Held out test set

Takeaway: Stock model misclassifies 86% of medium queries as complex. Our fine-tune fixes this.

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Impact & Close

One change. Benefits at every layer of the stack.

Business

60–98% cost savings on AI inference through intelligent right-sizing

60–98% saved

End Users

Simple queries get fast answers. Complex queries get powerful models.

Right model, always

Data Centers

Less unnecessary GPU load on premium clusters. Resources freed.

Less waste

Energy Grid

Up to 180× less energy per routed-away premium call. Compounding effect.

180× difference

Compliance

Contract constraints become routing rules. Enforced per request, automatically.

Zero human error

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> "ModelGate is the lowest-friction AI optimization available today. You change one line of code. We eliminate thousands of wasteful premium model calls automatically."

Thank you.

KSU · Social Good Hackathon · 2026

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